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## EMERGENCY APPARATUS AND SURGICAL INSTRUMENT


### Abstract

Disclosed are an emergency apparatus and a surgical instrument, which includes a transmission mechanism, a connection mechanism connected to the transmission mechanism, and an electric mechanism. The connection mechanism includes a first gear. The connection mechanism is disengaged from the electric mechanism via the first gear. The emergency apparatus includes: a movable shaft, which is movably disposed through the first gear. When the movable shaft and the first gear are in a first meshing state, the first gear is rotatable around the movable shaft. When the movable shaft and the first gear are in a second meshing state, the first gear is prevented from rotating around the movable shaft and can be driven to move by the movement of the movable shaft so as to disengage from the electric mechanism. The present application enables manual reset of the transmission mechanism after firing of the stapler.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/132808, filed on Nov. 21, 2023, which claims priority to Chinese Patent Application No. 202211474601.7, filed on Nov. 23, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
[0002] The present application relates to the technical field of medical instruments, and in particular, to an emergency apparatus and a surgical instrument.
BACKGROUND
[0003] In some conventional surgical instruments, an operation performed by surgeons is received at a proximal end by a firing mechanism, which drives a transmission shaft to move toward the distal end, thereby actuating an operating mechanism located at a distal end to perform corresponding surgical procedures. One example of such surgical instruments is a surgical stapler. Existing surgical stapler includes a stapler body, a firing mechanism (e.g., a firing handle) movably connected to the stapler body, and an end effector coupled to the stapler body. The stapler body includes a driving assembly. The end effector includes an anvil and a staple cartridge that are arranged opposite each other. During surgery, firstly, tissue is disposed between the anvil and the staple cartridge, the distance therebetween is then adjusted to gradually clamp the tissue, an operation received by the firing mechanism subsequently drives the driving assembly to move toward the distal end, enabling staples to form against the anvil. A stapling of tissue is thus completed.
[0004] During surgery, typically, surgeons are required to apply a significant amount of force to the firing mechanism, which presents inconvenience. To address this, a motor-driven surgical instrument has been developed. Once a motor thereof is activated, an output shaft of the motor rotates in a specific direction, so as to move the transmission mechanism toward the distal end. After the completion of one surgical procedure, the output shaft of the motor can be reversed, which is referred as an electric retraction function, and in response to which the transmission mechanism moves toward the proximal end to reset the entire instrument. However, malfunctions occurring in the motor, the associated mechanical/electronic components may impede the electric retraction function, preventing instruments from resetting, locking a jaw opening of the end effector, thus compromising surgical procedures.
SUMMARY
[0005] To address the aforementioned issues in prior art, the objective of the present application is to provide an emergency apparatus and a surgical instrument. In an event that an electric mechanism fails to reset a transmission mechanism after the completion the firing, an emergency apparatus is suitable for manually resetting the transmission mechanism.

[0006] In some embodiments of the present application, an emergency apparatus for a surgical instrument is provided. The surgical instrument includes: a transmission mechanism, driven to move toward or away from an end effector, of the surgical instrument; a connection mechanism connected to the transmission mechanism; and an electric mechanism, connected to the connection mechanism, configured to move the connection mechanism, thereby driving the transmission mechanism to move. The connection mechanism includes a first gear. The connection mechanism is configured to disengage from the electric mechanism in response to the disengagement of the first gear from the electric mechanism. The emergency apparatus includes a movable shaft movably disposed through the first gear, when the movable shaft and the first gear are in a first meshing state, the first gear is rotatable around the movable shaft; when the movable shaft and the first gear are in a second meshing state, the first gear is prevented from rotating around the movable shaft, and the first gear is configured to move in response to a movement of the movable shaft, thereby being disengaged from the electric mechanism, where the movement direction of the movable shaft is consistent with the movement direction of the first gear.
[0007] In some embodiments, when the movable shaft and the first gear are located at a first relative position, they are in the first meshing state; when the movable shaft and the first gear are located at a second relative position, they are in the second meshing state; where the movable shaft and the first gear convert from the first relative position to the second relative position in response to a movement of the movable shaft in a first direction. The first direction is perpendicular to a direction in which the transmission mechanism moves toward or away from the end effector of the surgical instrument.
[0008] In some embodiments, the movable shaft includes a first mating portion, a second mating portion and a third mating portion, while the first gear includes a fourth mating portion and a fifth mating portion. When the movable shaft and the first gear are in the first meshing state, the first mating portion cooperates with the fourth mating portion to allow the first gear to rotate around the movable shaft. When the movable shaft and the first gear are in the second meshing state, the second mating portion cooperates with the fourth mating portion to prevent the first gear from rotating around the movable shaft. When the movable shaft is driven to move, the third mating portion cooperates with the fifth mating portion to drive the first gear to move, thereby disengaging the first gear from the electric mechanism.
[0009] In some embodiments, the first gear includes a shaft hole through which the movable shaft is disposed. The movable shaft is provided with a first threaded portion. The emergency apparatus further includes a driving member having a second threaded portion for cooperating with the first threaded portion. When the driving member rotates, the movable shaft is driven to move.
[0010] In some embodiments, both ends of the movable shaft protrude beyond end faces at two sides of the first gear, respectively, and the first threaded portion is disposed at a first end of the movable shaft, the third mating portion being disposed at a second end thereof. The third mating portion includes an outer diameter greater than an inner diameter of the shaft hole of the first gear.
[0011] In some embodiments, the first mating portion of the movable shaft is located between the first threaded portion and the second mating portion and includes an outer diameter less than or equal to the inner diameter of the shaft hole of the first gear.
[0012] In some embodiments, in an event that the movable shaft moves in response to the rotation of the driving member after the first gear has disengaged from the electric mechanism, the first gear rotates in response to the movement of the movable shaft due to a cooperation formed between the second mating portion and the fourth mating portion.
[0013] In some embodiments, the emergency apparatus further includes an elastic member configured to apply a biasing force to the first gear in a second direction opposite to the first direction.
[0014] In some embodiments, the first threaded portion includes an external thread distributed on an outer surface of the movable shaft. The driving member is at least partially sleeved over the first

threaded portion. The second threaded portion includes an internal thread distributed on an inner surface of the driving member.

[0015] In some embodiments, the emergency apparatus further includes a first operating lever, the driving member is provided at a first end of the first operating lever.

[0016] In some embodiments, the first end of the first operating lever is provided with an mating hole or a mating groove. The driving member is at least partially embedded within said mating hole or mating groove.

[0017] In some embodiments, the outer surface of the driving member cooperates with an inner surface of the mating hole or the mating groove to form a non-rotatable connection.

[0018] In some embodiments, the driving member is detachably connected to the first end of the first operating lever.

[0019] In some embodiments, the emergency apparatus further includes a second operating lever connected to a second end of the first operating lever.

[0020] In some embodiments, the emergency apparatus further includes an outer frame, the first gear and the movable shaft are at least partially housed within the outer frame, the outer frame includes a receiving slot disposed on one side thereof, the second operating lever is at least partially housed within the receiving slot.

[0021] In some embodiments, the electric mechanism includes a motor and a motor gear assembly, an output shaft of the motor is configured to drive the motor gear assembly to rotate. The motor gear assembly is at least partially located on one side of the first gear along the second direction and includes a second gear, the second direction opposites to the first direction. When the electric mechanism is connected to the connection mechanism, the first gear meshes with the second gear, with the output shaft of motor parallel to the first direction.

[0022] In some embodiments, the motor gear assembly includes a first motor gear and a second motor gear that have axes arranged crossly. The output shaft of motor could be parallel to the first direction, or could be oriented at a first angle.

[0023] The present application further provides a surgical instrument including any of the aforementioned emergency apparatuses.

[0024] The emergency apparatus and surgical instrument provided in this application have the following advantages:

[0025] In an event that the electric mechanism fails to reset the transmission mechanism after the completion of instrument firing, the movable shaft is configured to drive the first gear to disengage from the electric mechanism, further of rotating the first gear to reset the transmission mechanism, thereby achieving an emergency manual reset. This enables a convenient operation under emergency, ensuring uninterrupted surgical procedures.

[0026] Furthermore, since the driving member provided at the first end of the first operating lever includes the second threaded portion cooperating with the first threaded portion of the movable shaft, the movable shaft moves in response to the rotation of the driving member. Such configuration guarantees a maintained connection between the first operating lever and movable shaft during an operation of the emergency apparatus, preventing the first lever form disengaging form the movable shaft, thereby facilitating the emergency manual reset. Such configuration further avoiding potential operational failures on emergency manual reset that might exacerbate surgeons' stress in emergency scenarios, thereby safeguarding surgical outcomes.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] Further features, objectives, and advantages of the present application will become more apparent from the detailed description of non-limiting embodiments with reference to the

accompanying drawings.
[0028] FIG. **1** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0029] FIG. **2** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0030] FIG. **3** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0031] FIG. **4** illustrates a structural schematic view of the outer frame according to the first embodiment of the present application.
[0032] FIG. **5** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame omitted.
[0033] FIG. **6** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame and the inner frame omitted.
[0034] FIG. **7** illustrates a front view of the emergency apparatus according to the first embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame and the inner frame omitted.
[0035] FIG. **8** illustrates a schematic view of the movable shaft according to the first embodiment of the present application, where the movable shaft cooperates with other instrument components.
[0036] FIG. **9** illustrates a structural schematic view of the operating mechanism according to the first embodiment of the present application, where the operating mechanism cooperates with the driving member.
[0037] FIG. **10** illustrates a structural schematic view of the movable shaft according to the first embodiment of the present application, where the movable shaft cooperates with the first gear.
[0038] FIG. **11** illustrates a schematic view of the movable shaft according to the first embodiment of the present application, where the movable shaft disengages from the first gear.
[0039] FIG. **12** illustrates a schematic view of the movable shaft according to the first embodiment of the present application, where the movable shaft disengages from the first gear.
[0040] FIG. **13** illustrates a schematic view of the emergency apparatus according to the first embodiment of the present application, where the first gear has disengaged from the electric mechanism, and the emergency apparatus cooperates with other instrument components.
[0041] FIG. **14** illustrates a schematic view of the emergency apparatus according to the second embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0042] FIG. **15** illustrates a schematic view of the emergency apparatus according to the second embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame omitted.
[0043] FIG. **16** illustrates a front view of the emergency apparatus according to the second embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame and the inner frame omitted.
[0044] FIG. **17** illustrates a schematic view of the emergency apparatus according to the second embodiment of the present application, where the emergency apparatus cooperates with other instrument components with the outer frame and the inner frame omitted.
[0045] FIG. **18** illustrates a bottom view of the movable shaft according to the second embodiment of the present application, where the movable shaft cooperates with other instrument components.
[0046] FIG. **19** illustrates a schematic view of the emergency apparatus according to the second

embodiment of the present application, where the first gear cooperates with the second motor gear, and the emergency apparatus cooperates with other instrument components.
[0047] FIG. **20** illustrates a schematic view of the emergency apparatus according to the second embodiment of the present application, where the first gear has disengaged from the second motor gear, and the emergency apparatus cooperates with other instrument components.
[0048] FIG. **21** illustrates a front view of the emergency apparatus according to the third embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0049] FIG. **22** illustrates a schematic view of the emergency apparatus according to the third embodiment of the present application, where the emergency apparatus cooperates with other instrument components.
[0050] FIG. **23** illustrates a front view of the emergency apparatus according to the third embodiment of the present application, where the first gear has disengaged from the electric mechanism, and the emergency apparatus cooperates with other instrument components.
[0051] FIG. **24** illustrates a structural schematic view of the operating mechanism according to the third embodiment of the present application.
DETAILED DESCRIPTIONS OF THE EMBODIMENTS
[0052] Exemplary embodiments will now be described more fully with reference to the accompanying drawings. However, the exemplary embodiments may be implemented in various forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided to make this application thorough and complete, and to fully convey the concepts of the exemplary embodiments to those skilled in the art. Throughout the drawings, like reference numerals designate identical or similar structures, and repetitive descriptions thereof will be omitted. The terms ‘or’ and ‘either’ in the specification may indicate ‘and’ or ‘alternatively’. Although terms such as ‘upper’, ‘lower’, and ‘between’ may be used herein to describe various exemplary features and elements of the present application, these terms are used merely for convenience, e.g., based on the orientation shown in the accompanying drawings. Nothing in this specification should be interpreted as requiring a specific three-dimensional orientation to fall within the scope of the present application. While terms such as ‘first’ to ‘fifth’ are used to denote certain features, they are merely indicative and do not limit the quantity or importance of the specific features.
[0053] The present application provides an emergency apparatus for a surgical instrument and a surgical instrument including said apparatus. The surgical instrument includes an instrument body and an end effector located at a distal end of the instrument body. The instrument body includes: a transmission mechanism that can be driven to move toward or away from the end effector of the surgical instrument; a connection mechanism connected to the transmission mechanism; and an electric mechanism connected to the connection mechanism and configured to drive the connection mechanism to move, thereby driving the transmission mechanism to move. The connection mechanism includes a first gear. The connection mechanism disengages from the electric mechanism in response to the disengagement of the first gear from the electric mechanism. The surgical instrument may be, for example, a surgical stapler or other types of surgical instruments.
[0054] The emergency apparatus includes: a movable shaft movably disposed through the first gear, when the movable shaft and the first gear are in a first meshing state, the first gear is rotatable around the movable shaft; when the movable shaft and the first gear are in a second meshing state, the first gear is prevented from rotating around the movable shaft, and the first gear is driven to rotate by the rotation of the movable shaft. The first gear is configured to move in response to a movement of the movable shaft, thereby being disengaged from the electric mechanism, and the movement direction of the movable shaft is consistent with the movement direction of the first gear. In an event that the electric mechanism fails to reset the transmission mechanism after the firing of instrument, the movable shaft is configured to drive the first gear to disengage from the

electric mechanism, and drive the first gear to rotate to reset the transmission mechanism, thereby achieving an emergency manual reset.

[0055] The structure of the emergency apparatus and the surgical instrument according to specific embodiments will be described in detail with reference to the drawings. It should be understood that these embodiments do not limit the protection scope of this application.

[0056] In the first embodiment of the present application, the surgical instrument comprises an instrument body and an end effector located at the distal end of the instrument body, with an emergency apparatus disposed at the instrument body. This embodiment is illustrated taking a surgical stapler as an example, where the end effector refers to the end effector performing stapling/cutting actions. FIGS. **1-13** illustrate the emergency apparatus according to the first embodiment, where the emergency apparatus cooperates with other instrument components including, but not limited to, a transmission mechanism **6**, a connection mechanism **4**, and an electric mechanism **5**.

[0057] As shown in FIGS. **1-8**, the instrument body includes a transmission mechanism **6** that can be driven to move toward or away from the end effector of the surgical instrument, a connection mechanism **4** connected to the transmission mechanism **6**, and an electric mechanism **5** connected to the connection mechanism **4** and configured to drive the connection mechanism **4** to move, thereby driving the transmission mechanism **6** to move. The transmission mechanism **6** is provided with a teeth surface **61** on its side facing the connection mechanism **4**. The connection mechanism **4** includes a first gear **41** meshing with the teeth surface **61** of the transmission mechanism **6**. The connection mechanism **4** is configured to disengage from the electric mechanism **5** in response to a disengagement of the first gear **41** from the electric mechanism **5**. The electric mechanism **5** comprises a motor **51** and a motor gear assembly including a second gear **52** disposed on the output shaft of the motor **51**. In an initial state, the first gear **41** meshes with the second gear **52**, and the motor **51** is configured to drive the transmission mechanism **6** to move to the distal end through the second gear **52**, or to move to the proximal end through the first gear **41**. When the first gear **41** is disengaged from the second gear **52**, the connection mechanism **4** decouples from the electric mechanism **5**, rendering the electric mechanism **5** incapable of driving the transmission mechanism **6**. The output shaft of the motor **51**, the axial direction of the second gear **52**, and the axial direction of the first gear **41** are all parallel to a first direction. In alternative embodiments, the connection mechanism **4** may include multiple first gears **41**, and/or, the motor gear assembly may comprise multiple second gears **52**. One or more first gears **41** mesh with one or more second gears **52** so as to transmit power from the motor **51** to the transmission mechanism **6**. Once those previously engaged gears are disengaged, the connection mechanism **4** and electric mechanism **5** become decoupled.

[0058] In the present application, the distal end and proximal end are defined relative to the operator, that is, one side closer to the operator is the proximal end, while another side farther from the operator (closer to the surgical site) is the distal end. An axial direction follows the longitudinal axis of the stapler, either from distal to proximal or proximal to distal. For example, in the perspective of FIG. **1**, direction S**1** represents a distal-to-proximal direction, and the transmission mechanism **6** may move along S**1** or its opposite direction. Direction S**2** in FIG. **1** is defined as a vertical direction (a height direction), while direction S**3** represents a transverse direction (a width direction). For any component, the inner side indicates the portion closer to its center, and the outer side indicates the portion farther from its center.

[0059] As shown in FIGS. **1-8**, the emergency apparatus comprises an outer frame **1**, an inner frame **7** and a movable shaft **43**. The outer frame **1** is sleeved over the first gear **41** and transmission mechanism **6**, while the inner frame **7** is positioned inside the outer frame **1**. The movable shaft **43** is movably disposed through the shaft hole **413** of the first gear **41**. The movable shaft **43** and the first gear **41** could form two meshing states. When the movable shaft **43** and first gear **41** are located at a first relative position, the movable shaft **43** and first gear **41** are in the first meshing state (in a state of FIG. **6**), the first gear **41** is rotatable around the movable shaft **43**. When the movable shaft **43** and first gear **41** are located at a second relative position, the movable shaft **43** and first gear **41** are in the second meshing state (in a state of FIG. **13**), the first gear **41** is prevented from rotating around the movable shaft **43**, the first gear **41** is rotatable in response to the rotation of the movable shaft **43**. While the movable shaft **43** moves along the first direction, the first gear **41** is driven to move along the same direction to be disengaged from the electric mechanism **5**. Specifically, the second gear **52** is at least partially positioned on one side of the first gear **41** in the second direction, and the second direction opposites to the first direction. When the first gear **41** is driven by the movable shaft **43** to move along the first direction, the first gear **41** and the second gear **52** become out of meshing, thereby disengaging the first gear **41** from the electric mechanism **5**.

[0060] In an event that an electric mechanism **5** fails to reset a transmission mechanism **6** after the firing of stapler, the manual reset applied to the transmission mechanism **6** could be achieved in two steps: Step 1: Driving the movable shaft **43** to move along the first direction with the first gear **41** remaining stationary, such that the movable shaft **43** and first gear **41** convert from the first relative position to the second relative position, thereby converting them from the first meshing state to the second meshing state, and continuously driving the movable shaft **43** to move along the first direction which drives the first gear **41** in the same direction to disengage from the electric mechanism **5** (being in a state as shown in FIG. **13**). Step 2: driving the movable shaft **43** to rotate, thus driving the first gear **41** in its second meshing state to rotate, whereupon the first gear **41** cooperates with the teeth surface **61** of the transmission mechanism **6** to drive the transmission mechanism **6** to move toward the proximal end. In this embodiment, the first direction corresponds to upward direction D**1** in FIG. **6**, parallel to direction S**2** in FIG. **1**, and is perpendicular to the movement direction of the transmission mechanism **6** (direction S**1** or its opposite direction).

[0061] The movable shaft **43** is primarily driven in two modes: In the Step 1, the movable shaft **43** is driven to move linearly along the first direction and in the Step 2, the movable shaft **43** is driven to rotate. In this embodiment, the emergency apparatus further includes an operating mechanism **2** and a driving member **3**, which are used for actuating the movable shaft **43** in these two modes. As shown in FIG. **2-6**, the operating mechanism **2** includes a first operating lever **21** and a second operating lever **22**. The driving member **3** is disposed at a first end of the first operating lever **21**. The second end of the first operating lever **21** is pivotally connected to the second operating lever **22** via a pivot shaft **23**. The operating mechanism **2** has two states: a storage state shown in FIGS. **2-3** where the operating mechanism **2** is folded to be accommodated on one side of the outer frame **1** with the second operating lever **22** partially received in the receiving slot **12** of the outer frame **1** shown in FIG. **4**. And an operational state where the driving member **3** located at the first operating lever **21** cooperates with the top portion of the movable shaft **43**, enabling the operation performed by the users on the second operating lever **22** sequentially transmit an input, through the first operating lever **21** and driving member **3**, to the movable shaft **43**, so as to drive the movable shaft **43** to move. The inner frame **7** is provided with a limiting portion **71**, sleeved over the top end of the movable shaft **43**, located beneath the driving member **3**. The limiting portion **71** limits a mating position between the first operating lever **21** and movable shaft **43**. In this embodiment, the first end of the first operating lever **21** is provided with a mating hole **211**, the driving member **3** is at partially embedded within the mating hole **211**. The first operating lever **21** and the driving member **3** may form a fixed connection, or alternatively, the driving member **3** may be removable from the mating hole **211** when the driving member **3** is not in use. An outer surface of the driving member **3** cooperates with an inner surface of the mating hole **211** and a non-rotatable connection is formed therebetween. In this embodiment, these two surfaces are matching polygonal profiles. In alternative implementations, the driving member **3** and first operating lever **21** could be integrated, or be fixed together via fasteners.

[0062] As shown in FIGS. **6-12**, the movable shaft **43** includes a first threaded portion **431**, a first mating portion **434**, a second mating portion **432**, and a third mating portion **433**, which are sequentially arranged along the axial direction of the movable shaft **43**. Both ends of the movable shaft **43** protrude beyond two end faces of the first gear **41**, respectively, with the first threaded portion **431** disposed at the first end of the movable shaft **43** and the third mating portion **433** formed as a stepped portion at the second end thereof. The driving member **3** is provided with a second threaded portion **31**. In this embodiment, the first threaded portion **431** includes an external thread while the second threaded portion **31** includes an internal thread. An outer diameter of the first mating portion **434** equals to or slightly exceeds the maximum outer diameter of the first threaded portion **431**. The second threaded portion **31** and the first threaded portion **431** can form a threaded connection. When the second threaded portion **31** cooperates with the first threaded portion **431**, the movable shaft **43** linearly moves along the first direction in response to the rotation of the driving member **3**. In an event that the movable shaft **43** to a position where the first mating portion **434** cooperates with the second threaded portion **31**, with these two form a non-rotatable connection, enabling the movable shaft **43** rotates in the same direction in response to the rotation of the driving member **3**.

[0063] As shown in FIGS. **11-12**, the first mating portion **434** constitutes a plain shaft section of the movable shaft **43**, while the second mating portion **432** is provided with a first anti-rotation structure on its outer surface. The first gear **41** includes a fourth mating portion **411** with a second anti-rotation structure and a fifth mating portion **412** formed by a bottom surface of the first gear **41**. The first anti-rotation structure and the second anti-rotation structure are polygonal profiles matched correspondingly.

[0064] The outer diameter of the first mating portion **434** is less than the minimum size of the inner diameter of the fourth mating portion **411**. Consequently, when the movable shaft **43** and first gear **41** are in the first meshing state, the first mating portion **434** cooperates with the fourth mating portion **411** to allow the first gear to rotate around the movable shaft **43**. The outer diameter of the third mating portion **433** is greater than the inner diameter of the shaft hole **413** of the first gear **41**. When the movable shaft **43** is driven to move along the first direction, the third mating portion **433** is prevented from entering the first gear **41**, abuts against and cooperates with the fifth mating portion **412**, so as to drive the first gear **41** to move along the first direction for disengaging the first gear **41** from the electric mechanism **5** without entering the first gear. When the first gear **41** disengages from the electric mechanism **5**, the movable shaft **43** and first gear **41** are in the second meshing state where the second mating portion **432** cooperates with the fourth mating portion **411**. The first anti-rotation structure cooperates with the second anti-rotation structure to prevent the first gear **41** from rotating around the movable shaft **43**, enabling that the movable shaft **43** drives the first gear **41** to rotate in the same direction.

[0065] As shown in FIGS. **11** and **12**, the emergency apparatus further includes an elastic member **42**, which is configured to apply a biasing force to the first gear **41** in a second direction opposite to the first direction. In this embodiment, the elastic member **42** includes a compression spring positioned above the first gear **41**. In alternative implementations, the elastic member **42** may include other elastic configurations such as an elastic pad, an elastic sheet positioned above the first gear **41**, or a tension spring positioned below the first gear **41**.

[0066] The operational procedures of the emergency apparatus will be illustrated in detail with reference to FIGS. **6** and **13**.

[0067] In an initial state where the stapler has not been fired, the first gear **41** simultaneously meshes with both the second gear **52** and a rack of the transmission mechanism **6**. The movable shaft **43** and first gear **41** are located at a first relative position where the first mating portion **434** of the movable shaft **43** cooperates with the fourth mating portion **411** of the first gear **41**, and thus the movable shaft **43** and first gear **41** are in the first meshing state where the first gear **41** and the movable shaft **43** are relatively rotatable. During the firing of staple, the motor **51** is started to drive the first gear **41** to rotate by means of the second gear **52** to propel the transmission mechanism **6** to move to the distal end, thereby actuating the end effector to perform cutting-stapling functions. During the firing of the stapler, the movable shaft **43** and first gear **41** are in the first meshing state and are relatively rotatable, resulting that the movable shaft **43** is prevented from rotating.

[0068] After the firing of the stapler, the emergency apparatus represents a state as shown in FIG. **6**, where the first gear **41** meshes with a teeth surface **61** of the transmission mechanism **6**, the teeth surface **61** is located at the proximal end of the transmission mechanism **6**. In an event that the electric mechanism or a relating mechanic structure fails, requiring a manual reset of the transmission mechanism, the aforementioned Step 1 and Step 2 are adopted to reset the transmission mechanism. Specifically, In Step 1, firstly cooperating the second threaded portion **31** of the driving member **3** with the first threaded portion **431** located at the top end of the movable shaft **43**, secondly operating the second operating lever **22** to rotate, transmitting such motion through the first operating lever **21** to the driving member **3**, rotating the driving member **3**, thereby moving the movable shaft **43** along the first direction, and the movable shaft **43** and the first gear **41** enters into the second relative position, where the second mating portion **432** of the movable shaft **43** enters the fourth mating portion **411** of the first gear **41** to form a relatively prohibited rotation therebetween, so that the movable shaft **43** and the first gear **41** enters the second meshing state. Simultaneously, the third mating portion **433** of the movable shaft **43** abuts against the fifth mating portion **412** of the first gear **41**, the movable shaft **43** continues to move along the first direction, and the movable shaft **43** drives the first gear **41** to move along the same direction by such cooperation between the third mating portion **433** and the fifth mating portion **412**. The first gear **41** disengages from the second gear **52** to disengage the connection mechanism **4** from the electric mechanism **5**, while the elastic member **42** is compressed to form an elastic deformation. In this way, the emergency apparatus enters into a state as shown in FIG. **13**. The elastic member **42** facilitates proper alignment of the movable shaft **43** with the first gear **41**, enabling a smoother entry of the second mating portion **432** into the fourth mating portion **411**.

[0069] After the connection mechanism **4** disengages from the electric mechanism **5**, the second threaded portion **31** cooperates with the first mating portion **434** of the movable shaft **43**. Accordingly, the length of the first threaded portion **431** equals to a stroke of the movable shaft **43** from its initial position to a position where the first gear **41** disengages from the second gear **52**. Step 2 is performed at this moment: continuously rotating the second operating lever **22** to rotate the second threaded portion **31** through the first operating lever **21**, the second threaded portion **31** drives, through movable shaft **43**, the first gear **41** to rotate in the same direction, and subsequently the first gear **41** drives the transmission mechanism **6** engaging with the first gear **41** to move along a axial direction of the stapler to the proximal end, thereby achieving a reset of the transmission mechanism **6**. The emergency apparatus represents a simple structure with convenient operation, effectively enabling manual reset of the transmission mechanism **6** in an event that the electric mechanism **5** fails to reset.

[0070] FIGS. **14-20** illustrate a mating structure between the emergency apparatus and other instrument components in the second embodiment of the present application. This embodiment differs from the first embodiment primarily in that the operating mechanism **2** in the first embodiment drives the movable shaft **43** to move on a top side of the outer frame **1**, and the operating mechanism **2** in the second embodiment drives the movable shaft **43** to move on one lateral side of the outer frame **1**. As shown in FIG. **15**, the inner frame **7** serves to support and fix the first gear **41** and movable shaft **43**, and the movable shaft **43** is disposed through the inner frame **7** such that the first threaded portion **431** remains exposed for cooperating with the driving member **3** of the operating mechanism **2**. FIGS. **16-18** illustrate that in this embodiment, the motor gear assembly **53** includes a first motor gear **531** and a second motor gear **532** that have axes arranged crossly. FIG. **17** specifically illustrates an axis X**1** of the first motor gear **531** (aligned with the output shaft of motor **51**) and an axis X**2** of the second motor gear **532**. It can be seen from FIG. **17** that X**1** and X**2** intersect each other. The first motor gear **531** is disposed over the output

shaft of the motor, which maintains an angular offset relative to the first direction. The second motor gear **532** incorporates a first teeth portion **5321** and second teeth portion **5322**. In the initial state, the first teeth portion **5321** meshes with the first motor gear **531**, while the second teeth portion **5322** meshes with the first gear **41**. When the motor **51** is started, the motor **51** drives the first gear **41** to rotate sequentially through the first motor gear **531** and second motor gear **532**.
[0071] As shown in FIGS. **18-20**, the first direction D**2** (parallel to a direction S**3** in FIG. **18**) is perpendicular to the movement direction of the transmission mechanism **6** (a direction S**1** in FIG. **1**). In the initial state, the second teeth portion **5322** of the second motor gear **532** meshes with the first gear **41**. In a state as shown in FIG. **18**, rotation of the second operating lever **22** drives the movable shaft **43** to move along the first direction D**2**, which in turn drives the first gear **41** to move in the same direction to disengage from the second teeth portion **5322** of the second motor gear **532**, achieving the state illustrated in FIG. **19**. The first teeth portion **5321** has an outer diameter smaller than that of the second teeth portion **5322**, thereby preventing interference with the first gear **41**. With the movable shaft **43** and the first gear **41** now in the second meshing state, the transmission mechanism **6** may be reset by driving the movable shaft **43** and the first gear **41** to rotate through the driving member **3**.
[0072] FIGS. **21-24** illustrate the structural cooperation between the emergency apparatus and other instrument components according to the third embodiment of the present application. This embodiment differs from the first embodiment primarily in that the driving member **3** is detachably connected to the first end of the first operating lever **21**. The first end of the first operating lever **21** is provided with a mating groove **212**, and the driving member **3** is at least partially embedded. As shown in FIG. **24**, the outer surface of the driving member **3** serves as a mating surface **32** that mates with the inner surface of the mating groove **212** to form a non-rotatable connection. In this embodiment, both the mating surface **32** of the driving member **3** and the inner surface of the mating groove **212** are polygonal surfaces. When stowing the operating mechanism **2** during non-use, the driving member **3** may remain engaged with the top end of the movable shaft **43**, maintaining the threaded connection between its internal second threaded portion and the first threaded portion **431**. In the initial state, the second gear **52** meshes with the first gear **41**. In a configuration as shown in FIG. **22**, rotation of the second operating lever **22** drives the movable shaft **43** along the first direction D**3**, consequently moving the first gear **41** in the same direction to disengage from the second gear **52** and achieve the state illustrated in FIG. **23**. With the movable shaft **43** and the first gear **41** now in the second meshing state, the transmission mechanism **6** may be reset by driving the movable shaft **43** and the first gear **41** to rotate through the driving member **3**.
[0073] The above description with reference to specific preferred embodiments provides further detailed explanation of the present application, but should not be construed as limiting the implementation of the present application solely to these illustrations. For those skilled in the art to which the present application pertains, various simple derivations or substitutions may be made without departing from the inventive concept of the present application, and all such modifications shall be considered within the protection scope of the present application.

## Claims

**1**. An emergency apparatus for a surgical instrument, wherein the surgical instrument comprises: a transmission mechanism, driven to move toward or away from an end effector of the surgical instrument; a connection mechanism, connected to the transmission mechanism; and an electric mechanism, connected to the connection mechanism and configured to drive the connection mechanism to move, thereby driving the transmission mechanism to move, the connection mechanism comprises a first gear, the connection mechanism is capable of disengaging from the electric mechanism, wherein, the emergency apparatus comprises: a movable shaft movably disposed through the first gear, wherein when the movable shaft and the first gear are in a first meshing state, the first gear is rotatable around the movable shaft; when the movable shaft and the first gear are in a second meshing state, the first gear is prevented from rotating around the movable shaft and is configured to move in response to a movement of the movable shaft so as to disengage from the electric mechanism, a movement direction of the movable shaft is consistent with a movement direction of the first gear.

**2**. The emergency apparatus according to claim 1, wherein when the movable shaft and the first gear are located at a first relative position, the movable shaft and the first gear are in the first meshing state; when the movable shaft and the first gear are located at a second relative position, the movable shaft and the first gear are in the second meshing state; and the movable shaft and the first gear are capable of converting from the first relative position to the second relative position in response to the movement of the movable shaft in a first direction, and the first direction is perpendicular to a direction in which the transmission mechanism moves toward or away from the end effector of the surgical instrument.

**3**. The emergency apparatus according to claim 2, wherein the movable shaft comprises a first mating portion, a second mating portion and a third mating portion; the first gear comprises a fourth mating portion and a fifth mating portion; when the movable shaft and the first gear are in the first meshing state, the first mating portion cooperates with the fourth mating portion to allow the first gear to rotate around the movable shaft; when the movable shaft and the first gear are in the second meshing state, the second mating portion cooperates with the fourth mating portion to prevent the first gear from rotating around the movable shaft; and when the movable shaft is driven to move, the third mating portion cooperates with the fifth mating portion to drive the first gear to move, thereby disengaging the first gear from the electric mechanism.

**4**. The emergency apparatus according to claim 3, wherein the first gear comprises a shaft hole, the movable shaft is disposed through the shaft hole of the first gear, and the movable shaft is provided with a first threaded portion; and the emergency apparatus further comprises a driving member provided with a second threaded portion cooperating with the first threaded portion, wherein the movable shaft is configured to move in response to a rotation of the driving member.

**5**. The emergency apparatus according to claim 4, wherein both ends of the movable shaft protrude beyond two end faces of the first gear, respectively, the first threaded portion is disposed at a first end of the movable shaft, the third mating portion is disposed at a second end of the movable shaft, and an outer diameter of the third mating portion is greater than an inner diameter of the shaft hole of the first gear.

**6**. The emergency apparatus according to claim 5, wherein the first mating portion of the movable shaft is located between the first threaded portion and the second mating portion, and an outer diameter of the first mating portion is less than or equal to the inner diameter of the shaft hole of the first gear.

**7**. The emergency apparatus according to claim 6, wherein, after the first gear is disengaged from the electric mechanism, when the movable shaft is driven to rotate by the rotation of the driving member, the first gear is configured to rotate in response to a cooperation between the second mating portion and the fourth mating portion.

**8**. The emergency apparatus according to claim 4, wherein the emergency apparatus further comprises an elastic member configured to apply a biasing force to the first gear in a second direction opposite to the first direction.

**9**. The emergency apparatus according to claim 4, wherein the first threaded portion comprises an external thread distributed on an outer surface of the movable shaft, the driving member is at least partially sleeved over the first threaded portion; the second threaded portion comprises an internal thread distributed on an inner surface of the driving member.

**10**. The emergency apparatus according to claim 9, wherein the emergency apparatus further comprises a first operating lever, the driving member is provided at a first end of the first operating lever.

**11**. The emergency apparatus according to claim 10, wherein the first end of the first operating lever is provided with a mating hole or a mating groove, and the driving member is at least partially embedded within the mating hole or the mating groove.

**12**. The emergency apparatus according to claim 11, wherein an outer surface of the driving member cooperates with an inner surface of the mating hole or the mating groove to form a non-rotatable connection.

**13**. The emergency apparatus according to claim 11, wherein the driving member is detachably connected to the first end of the first operating lever.

**14**. The emergency apparatus according to claim 10, wherein the emergency apparatus further comprises a second operating lever connected to a second end of the first operating lever.

**15**. The emergency apparatus according to claim 14, wherein the emergency apparatus further comprises an outer frame, the first gear and the movable shaft are both at least partially located inside the outer frame; one side of the outer frame is provided with a receiving groove, and the second operating lever is at least partially disposed within the receiving groove.

**16**. The emergency apparatus according to claim 4, wherein the electric mechanism comprises a motor and a motor gear assembly, an output shaft of the motor is configured to drive the motor gear assembly to rotate, the motor gear assembly is at least partially located on one side of the first gear in a second direction opposite to the first direction, the motor gear assembly comprises a second gear, and when the electric mechanism is connected to the connection mechanism, the first gear meshes with the second gear with the output shaft of the motor being parallel to the first direction.

**17**. The emergency apparatus according to claim 16, wherein the motor gear assembly comprises a first motor gear and a second motor gear, the first motor gear and the second motor gear have axes arranged crossly, the output shaft of the motor is parallel to the first direction or forms at an angle with the first direction.

**18**. A surgical instrument, comprising the emergency apparatus according to claim 1.

**19**. The surgical instrument according to claim 18, wherein when the movable shaft and the first gear are located at a first relative position, the movable shaft and the first gear are in the first meshing state; when the movable shaft and the first gear are located at a second relative position, the movable shaft and the first gear are in the second meshing state; and the movable shaft and the first gear are capable of converting from the first relative position to the second relative position in response to the movement of the movable shaft in a first direction, and the first direction is perpendicular to a direction in which the transmission mechanism moves toward or away from the end effector of the surgical instrument.

**20**. The surgical instrument according to claim 19, wherein the movable shaft comprises a first mating portion, a second mating portion and a third mating portion; the first gear comprises a fourth mating portion and a fifth mating portion; when the movable shaft and the first gear are in the first meshing state, the first mating portion cooperates with the fourth mating portion to allow the first gear to rotate around the movable shaft; when the movable shaft and the first gear are in the second meshing state, the second mating portion cooperates with the fourth mating portion to prevent the first gear from rotating around the movable shaft; and when the movable shaft is driven to move, the third mating portion cooperates with the fifth mating portion to drive the first gear to move, thereby disengaging the first gear from the electric mechanism.